

◆ 16.6 Evaluation and Decision Scoring Layer

16.6.1 Instantaneous Evaluation Function

$$f_i(t) = \frac{\nabla \Phi(\Psi_i(t))}{\overline{\mathcal{E}}(\Psi_i(t))} - 2\sigma^2 \nabla \ln P_i(t)$$

where:

- $\nabla \Phi(\Psi_i)$: goal-directed gradient
- $\overline{\mathcal{E}}(\Psi_i)$: normalized energy influence
- $\nabla \ln P_i$: probabilistic correction term

Thus:

$$f_i(t) \in L^{\infty}$$

16.6.2 Windowed Evaluation Integral

$$G_{i,u}^k = \int_{t_k}^{t_k + \Delta T} f_{i,u}(t) e^{-\lambda(t-t_k)} dt$$

with:

- $f_{\{i,u\}}(t)$ bounded
- $\lambda > 0$

Thus:

$G_{\{i,u\}}^k$ remains bounded.

16.6.3 Normalization and Scaling

$$\tilde{G}_{\{i,u\}}^k = \frac{G_{\{i,u\}}^k}{E_{\{\text{ref}\}}}$$

$$D_u(k) = \frac{1}{\sum_i W_i^k}, \tilde{G}_{\{i,u\}}^k$$

Thus:

decision scores are dimensionless and comparable.

16.6.4 PAC-Based Weighting

$$W_i^k = \frac{PAC_i^k}{\sum_j PAC_j^k + \epsilon_w}$$

$$\sum_i W_i^k = 1$$

Thus:

weights are normalized and numerically stable.

16.6.5 Decision Score Definition

$$D_u(k) = \frac{1}{\sum_{i=1}^N W_i^k} \tilde{G}_{i,u}^k$$

Thus:

decision scores are bounded and consistent across agents.

16.6.6 Goal Alignment

$$f_{i,u}(t) \supset \nabla \Phi(\Psi_i)$$

Thus:

decisions are aligned with the system objective.

16.6.7 Temporal Stability

$$|G_{\{i,u\}}^{k+1} - G_{\{i,u\}}^k| \leq L_G \Delta t$$

Thus:

no abrupt changes in decision evaluation.

16.6.8 Time Consistency

$G_{\{i,u\}}^k$ is computed over $[t_k, t_k + \Delta T]$

Thus:

evaluation remains synchronized with system dynamics.

16.6.9 Integration with System Components

- From Section 16.2: depends on Ψ and P

- From Section 16.3: consistent with energy dissipation
- From Section 16.5: weights depend on PAC

Thus:

the evaluation layer is fully integrated with system dynamics.

Closure Statement (Part VI)

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\boxed{
\text{The system produces bounded, normalized,
and goal-aligned decision scores.}
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SEKHEM-TSDD Theory, Mohamed Abd Elnaby,
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